

Question

Write a Java program to demonstrate multiple inheritance using interfaces.

Create two interfaces I1 and I2 containing the methods add() and mul() respectively. Implement both interfaces in class C1 and perform addition and multiplication.

Steps for Execution

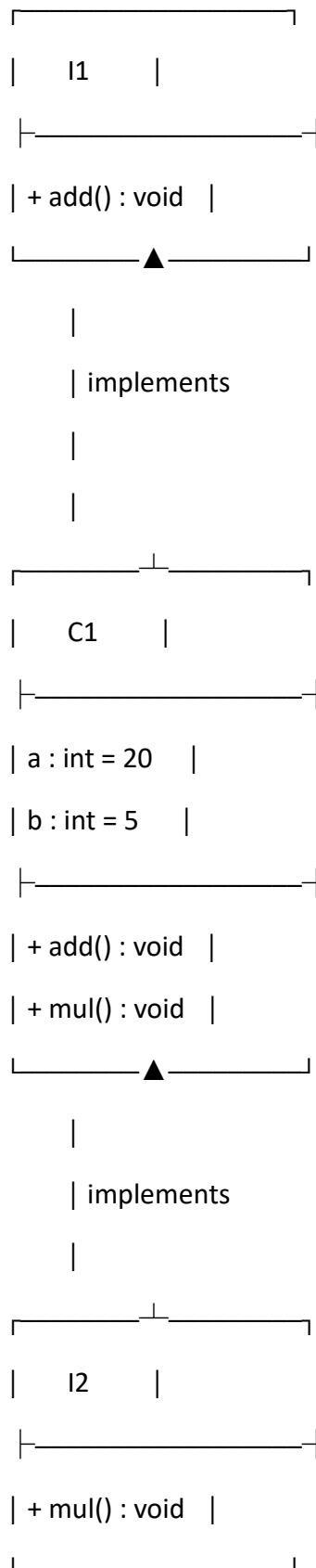
1. Create interface I1.
2. Declare the abstract method add() in I1.
3. Create interface I2.
4. Declare the abstract method mul() in I2.
5. Create class C1 and implement both I1 and I2.
6. Declare variables a = 20 and b = 5.
7. Implement add() to calculate a + b.
8. Implement mul() to calculate a * b.
9. Create the main() method inside the main class.
10. Create an object c of class C1.
11. Call c.add() and c.mul().
12. Display the results.

Algorithm

1. **Start**
2. Define interface I1 with method add().
3. Define interface I2 with method mul().
4. Define class C1 implementing I1 and I2.
5. Initialize a = 20 and b = 5.
6. Implement add() and calculate a + b.
7. Implement mul() and calculate a * b.
8. Create an object of C1.
9. Call add().
10. Call mul().
11. Display the results.

12. Stop

UML Class Diagram



This demonstrates **multiple inheritance through interfaces**, because C1 implements both I1 and I2.

Code

Java Multiple Inheritance Using Interfaces

```
package javapractice;
```

```
interface I1 {  
    void add();  
}
```

```
interface I2 {  
    void mul();  
}
```

```
class C1 implements I1, I2 {  
    int a = 20, b = 5;  
  
    @Override  
    public void add() {  
        System.out.println("Addition = " + (a + b));  
    }  
  
    @Override  
    public void mul() {  
        System.out.println("Multiplication = " + (a * b));  
    }  
}
```

```
public class MultipleInheritance {  
    public static void main(String[] args) {  
        C1 c = new C1();  
  
        c.add();  
        c.mul();  
    }  
}
```

Output

Addition = 25

Multiplication = 100